

Number line.

$a \in \mathbb{R}$ is visualized as a point on the line, located $|a|$ units from 0, accounting for sign.

2. Distance between real #s. $a \in \mathbb{R}, b \in \mathbb{R}$.

Def: distance between a and b is $|a - b|$.

3. Bounds on distances.

$$|a - b| \leq |a + (-b)| \leq |a| + |-b| = |a| + |b|.$$

Can this be sharp ($=$ instead of $\leq$)? Yes, if signs of a and b are opposite.

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0, \end{cases} \quad |xy| = |x||y|, \quad |x+y| \leq |x| + |y|, \quad |-x| = |x|.$$

3.1 Lower bound on $|a - b|$ in terms of $|a|, |b|$.

$$||a| - |b|| \leq |a - b| \quad - \text{reverse } \Delta \text{ inequality}$$

Pf of (x).

$$|a| = |a - b + b| \stackrel{\Delta}{\leq} |a - b| + |b|$$

$$1) \quad |a| - |b| \leq |a - b|$$

$$|b| = |b - a + a| \stackrel{\Delta}{\leq} |b - a| + |a| = |a - b| + |a|$$

$$2) \quad |b| - |a| \leq |a - b|$$

$$\Rightarrow (1), (2) \Rightarrow$$

$$||a| - |b|| \leq |a - b|$$

$$|x| = \max\{x, -x\}$$

Algebra of inequalities (for real #s).

1. If $a < b$ and $c < d$, then $a + c < b + d$.

2. If $a < b$ and $\lambda > 0$, then $\lambda a < \lambda b$ (!! if $\lambda < 0$, then $\lambda a > \lambda b$).

Also, note that $a < b$ and $c < d$ does *not* imply $ac < bd$.

3. Transitivity: if $a < b$ and $b < c$ then $a < c$.

Consider the putative: $a < b, c < d \Rightarrow ac < bd$ - not always true.

Ex. $a = 1, b = 2, c = -1, d = -\frac{2}{3}$.

Then $a < b, c < d$ but $ac = -1, bd = -\frac{4}{3}$.

Claim: If $0 < a < b$ and $0 < c < d$ then $ac < bd$.

Pf. Since $c > 0$, $a < b \Rightarrow ac < bc$ (mult. by a positive #).

Need $cb < bd$ to conclude by transitivity. Good if $b > 0$, but not in general.

As is, $b > 0$, $c < d \Rightarrow cb < bd$ (by a positive mult.).

And then

$$ac < cb < bd \Rightarrow ac < bd. \quad \square$$

Absolute value inequalities

1. $|x - a| < A$, $A > 0$. Solve.

Lemma $|x - a| < A \iff -A < x - a < A$. (1) (2)

Pf. Recall (def. of $|\cdot|$):

$$|x - a| = \begin{cases} x - a, & x - a \geq 0, \\ -(x - a) = a - x, & x - a < 0. \end{cases}$$

Case 1. Suppose $x - a \geq 0$. Then $|x - a| = x - a$.

(1) becomes $x - a < A$ and trivially $x - a \geq -A$ since $A > 0$, $-A < 0$.

Continue: $-A < x - a < A$.

Case 2. Suppose $x - a < 0$. Then (1) $\iff -(x - a) < A$ (multiply by -1 and reverse)
 $\Rightarrow x - a > -A$. Also, $x - a < 0$, $A > 0 \Rightarrow$ trivially $x - a < A$.

Combine: $-A < x - a < A$ $\square$

Solution set of $-A < x - a < A$ is $-A + a < x < A + a$ (the interval)



Variants (given) $|f(x) - g(x)| < A$, $A > 0$

$$-A < f(x) - g(x) < A \implies g(x) - A < f(x) < g(x) + A.$$

If f is invertible, then we can go further and solve for x if needed.

Def of f^{-1} (invertible). If $f(x_1) = f(x_2)$ then $x_1 = x_2$ for all $x_1, x_2 \in \mathcal{D}(f)$ (domain of f).

In practice, if f is continuous, may use $f \uparrow$ or $f \downarrow$.

$f \uparrow$: if $x_1 < x_2$ then $f(x_1) < f(x_2)$.

$f \downarrow$: if $x_1 < x_2$ then $f(x_1) > f(x_2)$. In that case the inverse function f^{-1} is well defined.

Thus

$$g(x) - A < f(x) < g(x) + A$$

$\implies$ (if f is increasing, then by def this implies)

$$f^{-1}(g(x) - A) < x < f^{-1}(g(x) + A).$$

(If f is decreasing, then we would have

$$f^{-1}(g(x) + A) < x < f^{-1}(g(x) - A).$$

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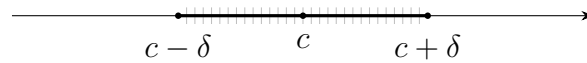
(Ex). Prove that $\lim_{x \rightarrow c} f(x) = f(c) \quad (L)$

$\forall \varepsilon > 0 \exists \delta > 0$ such that $0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

Steps. Consider $\{|f(x) - L| < \varepsilon \quad (x)\}$. Keep $\varepsilon > 0$ arbitrary.

Q: What is the set of x such that $(*)$ is true? Suppose this is the set $A(\varepsilon)$.

Q: does $A(\varepsilon)$ contain an interval $(c - \delta, c + \delta]$ for some $\delta(\varepsilon) > 0$?



If this is true, then $\delta(\varepsilon)$ is the prescription for finding δ given ε , and the proof is finished.

(Ex) Prove that $\lim_{x \rightarrow 3} x^3 = 27$.

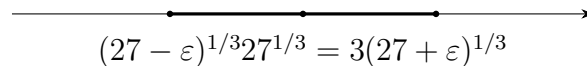
Consider $|x^3 - 27| < \varepsilon$. Can we solve for x ? (Always can solve for $f(x)$.)

$$|x^3 - 27| < \varepsilon \iff -\varepsilon < x^3 - 27 < \varepsilon$$

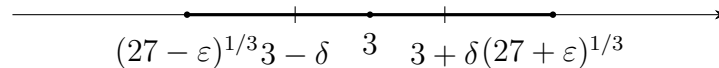
$$27 - \varepsilon < x^3 < 27 + \varepsilon$$

$f(x) = x^3$ is strictly increasing

$$\implies (27 - \varepsilon)^{1/3} < x < (27 + \varepsilon)^{1/3} \Rightarrow A(\varepsilon) \text{ is an interval } ((27 - \varepsilon)^{1/3}, (27 + \varepsilon)^{1/3}).$$



Picture.



δ is good.

$$\delta \leq \min \left\{ 3 - (27 - \varepsilon)^{1/3}, (27 + \varepsilon)^{1/3} - 3 \right\}.$$

This is a general prescription for finding δ given ε . $\square$

Q: $|f(x) - L| < \varepsilon \iff L - \varepsilon < f(x) < L + \varepsilon$ always.

What if we cannot solve for x algebraically? Then *estimate*.

Think of a simpler function $g(x) \geq 0$ such that

1. $|f(x) - L| \leq g(x)$ (at least on some fixed, non-zero length interval containing c);
2. $\lim_{x \rightarrow c} g(x) = 0$ (plausible);
3. $g(x) < \varepsilon$ is easier to solve.

(Ex). $\lim_{x \rightarrow 3} x^3 = 27$

$$|x^3 - 27| < \varepsilon.$$

$(x - 3)$ is of interest (to find good δ).

$$|x^3 - 27| = |x^3 - 3^3| = |(x - 3)(x^2 + 3x + 3^2)| = |x - 3| |x^2 + 3x + 3^2|.$$

Next: $(x^3 - 27) < \varepsilon \implies |x - 3| |x^2 + 3x + 3^2| < \varepsilon.$

If $x^2 + 3x + 3^2 \neq 0$,

$$|x - 3| < \frac{\varepsilon}{|x^2 + 3x + 3^2|}.$$

There is an interval I containing 3 such that

$$x^2 + 3x + 3^2 \geq C > 0 \quad \text{for all } x \in I.$$

If yes, then

$$|x - 3| < \frac{\varepsilon}{|x^2 + 3x + 3^2|} \leq \frac{\varepsilon}{C} \quad \forall x \in I.$$

Q. How to find I and C as above so that

$$x^2 + 3x + 3^2 \geq C > 0 \quad \forall x \in I?$$

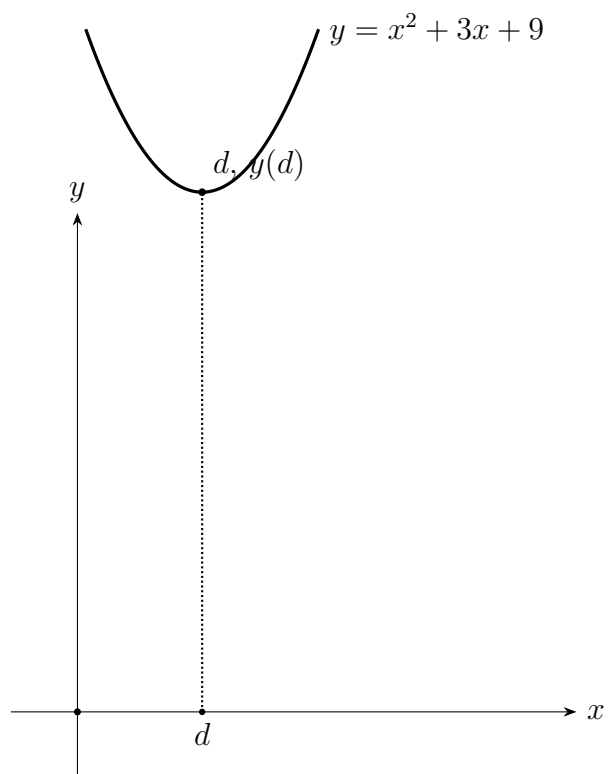
Coefficients 1, 3, 9. Discriminant

$$D = \sqrt{b^2 - 4ac} = \sqrt{3^2 - 4 \cdot 9} < 0 \implies \text{no real roots.}$$

$C = y$ -coordinate of the vertex of the parabola $y = x^2 + 3x + 3^2$.

d — the critical point of $y = x^2 + 3x + 3^2$.

$$\text{Inequality: } x^2 + 3x + 3^2 \geq d^2 + 3d + 3^2 \equiv C > 0 \quad \text{for all } x \in \mathbb{R}.$$



Now $|x - 3| < \frac{\varepsilon}{C}$. Therefore we can choose any $0 < \delta \leq \frac{\varepsilon}{C}$.

□